Illustration of using DIM Funtions to Compute Effect Size (R^2) in Multilevel Models

Overview

This example uses a simulated longitudinal dataset from a 12-month blood pressure management program (`bp_example.csv`). The dataset includes 300 adults who are scheduled to be assessed at baseline and at 3, 6, 9, and 12 months.

At each assessment, systolic blood pressure (`sbp`, mmHg) is measured, and participants report the number of days during the previous week on which they were physically active (`active_days`, ranging from 0 to 7). Participants' sex (`sex`) is included as a time-invariant variable. Time is coded from 0 to 4, with a 1 unit increase corresponding to one measurement occasion.

The simulated data include several features. On average, systolic blood pressure decreases over time, participants reporting more physically active days tend to have lower systolic blood pressure, and males have slightly higher systolic blood pressure on average. Individuals also differ in their initial blood pressure and rates of change over time. Attrition is introduced so that the number of observations decreases across occasions.

Variables

Variable	Description
`id`	Individual identifier
`time`	Measurement occasion coded 0, 1, 2, 3, and 4
`visit`	Visit label: Baseline, Month 3, Month 6, Month 9, Month 12
`active_days`	Number of physically active days during the previous week, ranging from 0 to 7
`sex`	Time-invariant individual characteristic: 1 = Female and 0 = Male
`sbp`	Systolic blood pressure in mmHg

Setup

```r
# Load functions
library(lme4)
library(nlme)

# Load the DIM functions.
source("r2lmer.R")
source("r2lme.R")
source("r2lmeRmat.R")

# Import data
bp_example<-read.csv("bp_example.csv")
```

Let's take a quick look at the data.

```r
head(bp_example)
```

```
##   id time sex    visit active_days   sbp
## 1  1    0   1 Baseline           0 147.1
## 2  1    1   1  Month 3           1 133.8
## 3  1    2   1  Month 6           2 132.7
## 4  1    3   1  Month 9           1 135.6
## 5  1    4   1 Month 12           3 139.9
## 6  2    0   0 Baseline           6 135.1
```

Number of observations at each time point:

```r
table(bp_example$time)
```

```
## 
##   0   1   2   3   4 
## 300 285 255 210 150
```

Number of males and females:

```
## 
##   0   1 
## 554 646
```

Summary of `sbp` and `active_days`

```r
summary(bp_example$sbp)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    94.4   130.5   138.1   137.6   145.0   173.7
```

```r
summary(bp_example$active_days)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   0.000   2.000   3.000   3.027   4.000   7.000
```

Overview of the examples

In the examples, we will fit increasingly complex longitudinal growth models using `lme4::lmer` and `nlme::lme` functions, and then showcase the application of `r2lmer()`, `r2lme()`, and `r2lmeRmat()` to compute R^2.

- Example 1: `r2lmer()` is illustrated with an `lmer()` model containing a random intercept and random slope of time.
- Example 2: `r2lme()` is illustrated with an `lme()` random intercept model and multiple fixed effects, including both time-varying and time-invariant predictors.
- Example 3: `r2lmeRmat()` is illustrated with an `lme()` model that includes random intercept and random slope of time, plus heteroscedastic and autocorrelated level-1 residuals.

Example 1: `r2lmer()`

The first model is estimated using `lme4::lmer()`. It includes time as a fixed effect and allows individuals to vary in both their initial systolic blood pressure (random intercept) and their rate of change over time (random slope).

```r
model_lmer <- lmer(
  sbp ~ time + (1 + time | id),
  data = bp_example
)

summary(model_lmer)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: sbp ~ time + (1 + time | id)
##    Data: bp_example
## 
## REML criterion at convergence: 8703.5
## 
## Scaled residuals: 
##     Min      1Q  Median      3Q     Max 
## -2.8086 -0.5466  0.0250  0.5300  3.5331 
## 
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr 
##  id       (Intercept) 67.65    8.225         
##           time         5.85    2.419    -0.18
##  Residual             44.40    6.663         
## Number of obs: 1200, groups:  id, 300
## 
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) 140.6022     0.5691 247.077
## time         -1.8785     0.2188  -8.583
## 
## Correlation of Fixed Effects:
##      (Intr)
## time -0.399
```

Extract target effect size

In our function, we use effectf and effectr to select fixed and random effects, separately.

effectf: The selected fixed effect terms for which the R^2 statistic is computed. Set `effectf <- NULL` if no fixed effect is selected.

effectr: The selected random effect terms for which the R^2 statistic is computed. Set `effectr <- NULL` if no random effect is selected. Set `effectr <- “(Intercept)”` to select the random intercept.

Selected fixed effects

To obtain the R^2 associated with the fixed effect of time:

```r
r2lmer(
  model_lmer,
  effectf = "time",
  effectr = NULL
)
```

```
## $`All Fixed Effects in the model`
## [1] "(Intercept)" "time"       
## 
## $`All Random Effects in the model`
## [1] "(Intercept)" "time"       
## 
## $`Variance partitioning`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   6.40921956  83.17190290  44.40110432 133.98222678   0.04783634   0.62076818 
##   R2Residual 
##   0.33139548 
## 
## $`Selected Fixed Effects`
## [1] "time"
## 
## $`Selected Random Effects`
## NULL
## 
## $`R2 for selected effects`
##     TotalVar    EffectVar           R2 
## 133.98222678   6.40921956   0.04783634 
## 
## $`Warning for correlated terms (Fixed)`
## [1] NA
## 
## $`Warning for correlated terms (Random)`
## [1] NA
```

This example illustrates the use of `r2lmer()` when the fitted model contains both a random intercept and a random time slope.

Example 2: `r2lme()`

The second model is estimated using `nlme::lme()`. It includes time, weekly physical activity, and sex as fixed effects, with a random intercept for individuals.

```r
model_lme <- lme(
  fixed = sbp ~ time + active_days + sex,
  random = ~ 1 | id,
  data = bp_example
)

summary(model_lme)
```

```
## Linear mixed-effects model fit by REML
##   Data: bp_example 
##        AIC      BIC    logLik
##   8763.701 8794.221 -4375.851
## 
## Random effects:
##  Formula: ~1 | id
##         (Intercept) Residual
## StdDev:    8.238619 7.492833
## 
## Fixed effects:  sbp ~ time + active_days + sex 
##                 Value Std.Error  DF   t-value p-value
## (Intercept) 144.19231 0.9857272 898 146.28014  0.0000
## time         -1.58312 0.1851658 898  -8.54975  0.0000
## active_days  -0.90677 0.2200014 898  -4.12167  0.0000
## sex          -2.48392 1.0597486 298  -2.34387  0.0197
##  Correlation: 
##             (Intr) time   actv_d
## time        -0.013              
## active_days -0.555 -0.411       
## sex         -0.583 -0.005  0.009
## 
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max 
## -3.28967858 -0.58158723  0.01629142  0.53649217  3.76207866 
## 
## Number of Observations: 1200
## Number of Groups: 300
```

Selected fixed effects

To obtain the R^2 associated with the the selected fixed effects of time, physical activity and sex:

```r
r2lme(
  model_lme,
  effectf = c("time", "active_days","sex"),
  effectr = NULL
)
```

```
## $`All Fixed Effects in the model`
## [1] "(Intercept)" "time"        "active_days" "sex"        
## 
## $`All Random Effects in the model`
## [1] "(Intercept)"
## 
## $`Variance partitioning`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   9.45810640  67.87483698  56.14254042 133.47548380   0.07086025   0.50851913 
##   R2Residual 
##   0.42062062 
## 
## $`Selected Fixed Effects`
## [1] "time"        "active_days" "sex"        
## 
## $`Selected Random Effects`
## NULL
## 
## $`R2 for selected effects`
##     TotalVar    EffectVar           R2 
## 133.47548380   9.45810640   0.07086025 
## 
## $`Warning for correlated terms (Fixed)`
## [1] NA
## 
## $`Warning for correlated terms (Random)`
## [1] NA
```

Selected fixed and random effects

To obtain the R^2 associated with the selected fixed effects together with the individual random intercept:

```r
r2lme(
  model_lme,
  effectf = c("time", "active_days","sex"),
  effectr = "(Intercept)"
)
```

```
## $`All Fixed Effects in the model`
## [1] "(Intercept)" "time"        "active_days" "sex"        
## 
## $`All Random Effects in the model`
## [1] "(Intercept)"
## 
## $`Variance partitioning`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   9.45810640  67.87483698  56.14254042 133.47548380   0.07086025   0.50851913 
##   R2Residual 
##   0.42062062 
## 
## $`Selected Fixed Effects`
## [1] "time"        "active_days" "sex"        
## 
## $`Selected Random Effects`
## [1] "(Intercept)"
## 
## $`R2 for selected effects`
##    TotalVar   EffectVar          R2 
## 133.4754838  77.3329434   0.5793794 
## 
## $`Warning for correlated terms (Fixed)`
## [1] NA
## 
## $`Warning for correlated terms (Random)`
## [1] NA
```

Fixed effect of time-invariant variable

Because sex is time invariant, this model also provides a simple illustration of R^2 for a between-individual fixed effect:

```r
r2lme(
  model_lme,
  effectf = "sex",
  effectr = NULL
)
```

```
## $`All Fixed Effects in the model`
## [1] "(Intercept)" "time"        "active_days" "sex"        
## 
## $`All Random Effects in the model`
## [1] "(Intercept)"
## 
## $`Variance partitioning`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   9.45810640  67.87483698  56.14254042 133.47548380   0.07086025   0.50851913 
##   R2Residual 
##   0.42062062 
## 
## $`Selected Fixed Effects`
## [1] "sex"
## 
## $`Selected Random Effects`
## NULL
## 
## $`R2 for selected effects`
##    TotalVar   EffectVar          R2 
## 133.4754838   1.5346747   0.0114978 
## 
## $`Warning for correlated terms (Fixed)`
## [1] "The selected fixed effect term(s) are correlated with other terms."
## 
## $`Warning for correlated terms (Random)`
## [1] NA
```

When selecting fixed effect for sex, the output shows a warning that “`Warning for correlated terms (Fixed)`:The selected fixed effect term(s) are correlated with other terms.”, therefore, the selected R^2 is not interpretable.

Together, these examples illustrate how `r2lme()` can be used to evaluate different combinations of selected fixed and random effects from the same fitted model.

Example 3: `r2lmeRmat()`

The third model uses a more complex level-1 residual covariance structure. In addition to random intercepts and random slope of time, the model specifies an ARH(1) residual correlation structure and allows residual variance to differ across measurement occasions.

```r
model_lmeRmat <- lme(
  fixed = sbp ~ time+active_days+sex,
  random = ~ 1 + time | id,
  correlation = corAR1(
    form = ~ time | id
  ),
  weights = varIdent(
    form = ~ 1 | time),
    control = lmeControl(
    opt = "optim"
  ),
  data = bp_example
)

summary(model_lmeRmat)
```

```
## Linear mixed-effects model fit by REML
##   Data: bp_example 
##        AIC      BIC    logLik
##   8599.512 8665.639 -4286.756
## 
## Random effects:
##  Formula: ~1 + time | id
##  Structure: General positive-definite, Log-Cholesky parametrization
##             StdDev    Corr  
## (Intercept) 7.1742685 (Intr)
## time        0.7045959 0.158 
## Residual    6.6540763       
## 
## Correlation Structure: AR(1)
##  Formula: ~time | id 
##  Parameter estimate(s):
##       Phi 
## 0.3771994 
## Variance function:
##  Structure: Different standard deviations per stratum
##  Formula: ~1 | time 
##  Parameter estimates:
##        0        1        2        3        4 
## 1.000000 1.014717 1.083464 1.484265 1.848650 
## Fixed effects:  sbp ~ time + active_days + sex 
##                 Value Std.Error  DF   t-value p-value
## (Intercept) 144.27909 0.9105273 898 158.45663   0.000
## time         -1.47151 0.2236116 898  -6.58066   0.000
## active_days  -1.01649 0.1903187 898  -5.34099   0.000
## sex          -2.30659 1.0236479 298  -2.25330   0.025
##  Correlation: 
##             (Intr) time   actv_d
## time        -0.058              
## active_days -0.519 -0.306       
## sex         -0.609 -0.006  0.011
## 
## Standardized Within-Group Residuals:
##         Min          Q1         Med          Q3         Max 
## -2.54375905 -0.58044932  0.01452345  0.56037719  2.93436588 
## 
## Number of Observations: 1200
## Number of Groups: 300
```

All fixed and random effects

To obtain the R^2 associated with the fixed effect of time, the random intercept and random slope of time:

```r
r2lmeRmat(
  model_lmeRmat,
  effectf = c("time","active_days","sex"),
  effectr = c("(Intercept)", "time")
)
```

```
## $`All Fixed Effects in the model`
## [1] "(Intercept)" "time"        "active_days" "sex"        
## 
## $`All Random Effects in the model`
## [1] "(Intercept)" "time"       
## 
## $`Level-1 residual covariance structure`
## [1] "ARH1"
## 
## $`Number of observations at each time point`
## [1] 300 285 255 210 150
## 
## $`Variance partitioning`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   9.12300174  56.48833485  78.14040208 143.75173868   0.06346359   0.39295758 
##   R2Residual 
##   0.54357883 
## 
## $`Variance partitioning (weighted residual variance)`
##     AllFixed    AllRandom     Residual     TotalVar   R2AllFixed  R2AllRandom 
##   9.12300174  56.48833485  68.92625971 134.53759631   0.06781005   0.41987025 
##   R2Residual 
##   0.51231969 
## 
## $`Selected Fixed Effects`
## [1] "time"        "active_days" "sex"        
## 
## $`Selected Random Effects`
## [1] "(Intercept)" "time"       
## 
## $`R2 for selected effects`
##    TotalVar   EffectVar          R2 
## 143.7517387  65.6113366   0.4564212 
## 
## $`R2 for selected effects (weighted residual variance)`
##    TotalVar   EffectVar          R2 
## 134.5375963  65.6113366   0.4876803 
## 
## $`Warning for correlated terms (Fixed)`
## [1] NA
## 
## $`Warning for correlated terms (Random)`
## [1] NA
```

```
## lmeWarning: lme() reported the following when fitting this model. Check the model before interpreting the resu
lts.
## Non-positive definite approximate variance-covariance
```

Because the simulated data have both unequal sample sizes across occasions and different residual variability over time. The weighted and unweighted R^2 calculations are not identical.

The output also shows a warning message that produced by the `nlme()` function: Non-positive definite approximate variance-covariance. Therefore, the R^2 should be interpret with caution.

These three examples use the simulated dataset to demonstrate different applications of the proposed DIM functions.